Milwaukee School of Engineering
PHY 1110

Introductory Physics

Group Problem Set 17: Work, Energy and Power

Fill in the following table characterizing the different forms of energy.

Energy Class	**Definition**	**Formula**
Kinetic Energy (K)		
Potential Energy (U)		
Gravitational Potential Energy (U_g) (near the earth)		

Conservation of Energy: Energy is conserved when conserved forces are responsible for change in motion.

1. What is a conserved force? Give an example?

2. What does it mean to be path-independent? Give examples of when path independence and path dependence are present and contrast the differences in circumstances (forces, setups).

3. List the forces that are generally conserved and associate them with either kinetic, potential or gravitational energy.

The conservation of energy dependent only on conserved forces is:

$$K_i + U_i = K_f + U_f$$

You will readily come across situations where an object experiences both conserved (gravity, spring) and non-conserved (tension, normal, friction, pushing/pulling) forces. The total mechanical energy of a system for an object experiencing both classes of forces is:

$$E_{init} + W_{non-conserve} = E_{final}$$

Solving conservation of energy problems can be performed by modifying the problem solving steps we've used in the past:

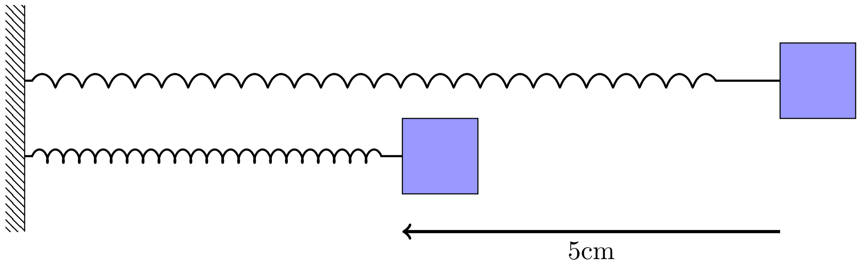
- Sketch a picture. Define beginning and end points and label them.

- Consider the forces involved in the problem. Label the conserved forces (gravity and spring). Look at the non-conservative forces (tension, normal, friction, pull/push) and label the forces that do work.

- Choose your axes and define your start and finish position values with respect to this. It helps to make either the beginning or end point the origin in at least one direction. For springs oriented vertically, it helps to set the relaxed spring position at y=0 (this will save you work).

- Set up the Work Energy Theorem: $E_{init} + W_{non-conserve} = E_{final}$. Each value ($E_{init}$ etc.) is represents the total energy of the system at those times/locations. This means that we include the energies associated with each of the bodies involved in the problem, sum those and then apply the Work Energy Theorem.

1. Use the following information to investigate the kinetic energy of springs.



A block (top block) sits on a table and is compressed against a spring (bottom block). The block has a mass of 100 g, the spring is compressed 5 cm along the table and the associated spring constant is 400 N/m. Assume that there is no air resistance or friction. The block is held in place compressing the spring until a time t=0 when it is released and the spring pushes the block back toward it's initial position (top block/spring). At some point (5cm from the release) the block will leave the spring and continue in the forward direction. .

(a) What is the velocity of the block at 5 cm away from the point it is initially released? (Hint: At this point, all potential energy has translated to kinetic energy).

(b) This velocity is called the muzzle velocity. We usually associate muzzle velocity with guns/riffles. In this case, muzzle velocity is the velocity corresponding to a moving object free of contact forces. If the table is 1 meter off the ground, and the block leaves the table 5 cm after it is released from its resting position, what is $\vec{v_o}$ of the block as it flies off the edge of the table? Draw a diagram showing the block at the height of the table, its initial velocity and probable trajectory.

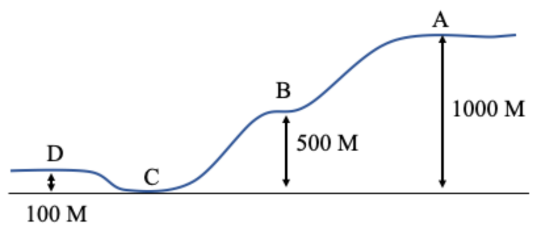
(c) Because there is no air resistance, we can use the kinematic equations of motion to determine the time of flight of the block. How long does it take for the block to fall to the floor?

(d) How far does the block travel horizontally from the base of the table?

(e) Use conservation of energy to find the velocity of the block right before it hits the ground.

(f) What is the velocity required to make the block travel twice as far? How much would the spring need to be compressed to achieve this velocity?

2. Answer the following questions about a 1000 kg car rolling down a hill (points A –¿ D) following the trajectory below. The velocity of the car at point A is 10 m/s. Assume the car is rolling without friction.



(a) Find the velocity of the car at point B assuming no friction.

(b) Find the velocity of the car at point C assuming no friction.

(c) Now, let's assume there is friction. If we know that the velocity of the car is 5 m/s at point B, how much work is being done by friction?

(d) If the distance driven from points A to B is 10,000 m, and the normal force can be approximated as mass*gravity, what is the coefficient of friction on the road?

(e) If the velocity of the car is 10 m/s at point B, how much work is being done by friction?

(f) If the velocity does not change between A and B, what can you say about the work done by gravity in comparison to the work done by friction?

(g) What is the coefficient of friction where the velocity at B is 0 m/s?

3. Now it's time for you to design your own problem. Choose a commonly used trip (to the coffee shop, class, Burrito Union) and rewrite problem 3 to reflect that trip. Be sure that the trip changes heights and has at least 2 stops. Now use conservation of energy to determine the amount of work generated by friction that helps or hinders you getting you through the trip.